Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/17/2012
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Crystal Growth of II-VI Compound Semiconductors		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ching.h.su@nasa.gov	**Fax:**	FY 256-544-8762
PI Organization Type:	NASA CENTER	**Phone:**	256-544-7776
Organization Name:	NASA Marshall Space Flight Center		
PI Address 1:	Mail Code EM31		
PI Address 2:	Materials and Processing Laboratory		
PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2006	**End Date:**	09/30/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Change in end date per P. Downey/MSFC (Ed., 9/28/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Two reasons have been put forward to account for the improved crystal quality when crystals are grown in a microgravity environment. The first is weight-related reductions in crystal strain and defects. These are thought to be caused by the weight of the crystals during processing at elevated temperatures and retained on cooling, particularly for materials with low yield strength. The second and more general reason is related to the reduction in density-gradient driven convection in the fluid during crystal growth in reduced gravity. The objectives and experiment description for the proposed flight experiment “Growth of Ternary Compound Semiconductors” by (1) vapor growth and (2) melt growth are described below.
1a) Vapor Growth

The physical vapor transport (PVT) crystal growth process consists of essentially three processes: sublimation of source

Task Description:

material, transport of vapor species and condensation of vapor species to form crystal. The latter two processes can be affected by the convection caused by gravitational accelerations on Earth. Reductions in such convection in low gravity is expected to yield a nearly diffusion-limited growth condition which results in more uniform growth rates (on the microscopic scale) and hence greater crystalline perfection and compositional homogeneity. This reduction of convective contamination in a reduced gravity environment will help to simplify the complexity of the coupled mass transport and growth kinetics problem. An improved comparison between the experimental results and theoretical simulations is also expected and will be beneficial to any growth process involving vapor mass transport or vapor-crystal interface growth kinetics.

The materials to be studied in this project are ZnSe and the related ternary compound semiconductors such as ZnSeTe, ZnSeS and ZnCdSe. The scientific objectives of the growth of ZnSe crystals in reduced gravity using physical vapor transport (PVT) processes are:

• to establish the relative contributions of gravity-driven fluid flows to (i) the non-uniform incorporation of impurities and defects and (ii) the deviation from stoichiometry observed in the grown crystals as a result of buoyancy-driven convection and irregular fluid-flows caused by growth interface fluctuations and interface evolution.

• to assess self-induced strain effects developed during processing at elevated temperatures and retained on cooling caused by the weight of the crystals from x-ray topography and by measuring the density and distribution of various crystalline defects.

• to obtain high quality space-grown materials for various thermophysical and electrical properties measurements and as substrates for device fabrication and thus assess device performance as influenced by a substantial reduction in gravity-related effects.

1b) Melt Growth

The melt growth of homogeneous ternary compound semiconductors, such as cadmium zinc telluride (Cd1-xZnxTe, $0 < x < 1$) is difficult because the thermal and compositional gradients in the melt just ahead of the growth interface during directional solidification can cause substantial gravity-driven fluid flows that alter the relative distribution of the two (CdTe and ZnTe) binary components and introduce structural defects into the grown crystals. The scientific objectives of the growth of CdZnTe crystals in reduced gravity using directional solidification process are:

• to investigate quantitatively the influences of gravitationally dependent phenomena on the growth process and the incorporation of impurities and defects in the resulting crystals.

• to better understand and control the growth parameters in the production process on Earth so as to increase the yield of usable material from the grown ingot.

• to assess strain effects developed during processing at elevated temperatures and retained on cooling, caused by the weight of the crystals and, possibly, the contact with the growth container, from the structural characterization and by measuring the density and distribution of various crystalline defects.

• to obtain high quality space-grown materials for various electrical and optical property measurements and as materials for the x-ray and gamma-ray detector fabrication as well as substrates for infrared device application and thus assess device performance as influenced by a substantial reduction in gravity-related effects.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

• In crystal growth there is a need to understand the relation between processes in the fluid phase (both liquid and vapor), such as buoyancy driven convection and the incorporation of impurities and defects in the resulting crystal
• Relation between fluid phase processes and the generation of defects in a grown crystal is an outstanding problem in materials growth

• Crystal quality greatly influences important electronic properties in materials • Ternary compound semiconductors are of vital national interest as sensors in x-ray telescopes and for homeland security, and as substrate materials for infrared sensors

• Studies in microgravity will be compared with modeling and will be used to optimize ground-based experiments

We have the following progresses on the categories of melt growth and the vapor growth of ternary II-VI compound semiconductors:
1. Melt growth:

• Crystal growth and characterization of CdTe grown by vertical gradient freeze. Crystals of CdTe were grown from melts by the unseeded vertical gradient freeze method. The thermal history of the melt and the thermal gradient at the growth interface were adjusted to optimize the crystalline quality of the grown crystals. The grown crystals were studied by various characterization techniques, including synchrotron white beam X-ray topography (SWBXT), chemical analysis by glow discharge mass spectroscopy (GDMS), low temperature photoluminescence (PL) and Hall measurements. The SWBXT images taken from various angles show grains with inhomogeneous strains, nearly strain-free grains, as well as twinning nucleated in the shoulder region of the boule. The GDMS chemical analysis shows the contamination of Ga at a level of 3900 ppb, atomic. The low temperature PL measurement exhibits the characteristic emissions of a Ga-doped sample. The Hall measurements show a resistivity of 1×107 ohm cm at room temperature and 3×109 ohm cm at 78K with the respective hole and electron concentration of 1.7×109 and 3.9×107 cm-3 at room temperature.

• Increase the grown CdZnTe boule size from 40 to 70mm diameter. More than ten crystals of CdZnTe with a diameter of 70mm have been processed by directional solidification with controlled Cd over pressure. The structural quality and the electrical resistivity were characterized and the results were comparable to that of the processed crystals with 40mm diameter.

• Melt growth of high-resistivity CdZnTe crystals by controlling Cd over-pressures. Cd1-xZnxTe crystals, with x = 0.20,

Task Progress:	were grown unseeded by closed-ampoule vertical directional solidification (Bridgman) technique under controlled Cd over-pressures. CdZnTe crystals of high electrical resistivity have been reproducibly obtained with the In dopant concentration of 4–6 ppm, atomic and a Cd reservoir temperature between 785 and 825 oC. A combination of three process modifications have contributed to the accomplishment: (1) homogenizing the ampoules of starting materials under the vacuum environment to minimize contamination, (2) performing growth under controlled Cd over-pressures with the optimal In dopant, and (3) controlling post-growth cooling procedures to maintain uniformity of electrical properties over the crystal. 2. Vapor growth: • Growth and characterization of ZnSe crystals by vapor transport. ZnSe bulk crystals were grown by self-seeded and seeded physical vapor transport technique in horizontal and vertical configurations. The impurities and defects in the grown crystals were studied by glow discharge mass spectroscopy (GDMS) and low-temperature photoluminescence (PL) measurements. The PL results on the starting material and the grown crystals are consistent with the low impurity levels measured by GDMS. The crystalline quality of the grown crystals was examined by synchrotron white beam X-ray topography (SWBXT) and high-resolution triple X-ray diffraction (HRTXD). The SWBXT shows that, aside from twins, the overall crystalline quality of the vapor-grown ZnSe crystals, especially in the contactless grown region, was quite high. The HRTXD results are in line with the SWBXT findings. The comparison between the HRTXD on a chemical-mechanically polished and a cleaved surface seems to indicate that polishing damage can obscure the true microstructure in the as-grown ZnSe crystals. • Establishing energy band Gap, intrinsic carrier concentration and Fermi level of CdTe bulk crystal between 304 K and 1067K. Optical transmission measurements were performed on CdTe bulk single crystals. It was found that when sliced and polished CdTe wafers were used, a white film started to develop on the sample surface and the wafer became opaque when it was heated above 530 K. Therefore, a bulk crystal of CdTe was first grown in the window area by physical vapor transport. The optical transmission was then measured between 304 and 1067 K and from which the energy band gap was derived. The band gaps of CdTe can fit well as a function of temperature by the Varshni expression. Using the band gap data, the high temperature electron-hole equilibrium was numerically calculated by assuming Kane's conduction band structure and a light-hole and a heavy-hole parabolic valence bands. The calculated intrinsic carrier concentrations agree well with the experimental data previously reported. The calculated intrinsic Fermi levels between 200 and 1200 K were also presented. • Measurements of thermal expansion coefficient of ZnSe crystal between 17 and 1080 °C. ZnSe is a semiconductor with zinc-blend (cubic) structure, under normal conditions, which has a wide transmission range covering the IR range of 0.6 to 20 µm. High optical quality ZnSe crystal has been used to manufacture optical components (windows, mirrors, lenses etc.) for high power IR lasers as well as etalons for IR spectrometers. The thermal expansion of ZnSe has been measured by a capacitance dilatometer for temperatures below 90 K and between 0 and 20 °C. The thermal expansion coefficient was determined to be 6.8×10-6 °C-1 at 10 °C. The method needed a calibrated dilatometer cell with known values of thermal expansion in the measured temperature ranges. Other data reviewed gives a value of 7.2×10-6 °C-1 at the temperature of 300 K. To measure the thermal expansion of ZnSe at elevated temperature, e.g., above 1000 °C, the sample has to be sealed inside an ampoule to prevent the sublimation and oxidization of the ZnSe sample, which excludes the dilatometry method and makes optical methods, such as interferometry, the only possibility. The Michelson interferometer usually introduces large thermal errors because of the relative spatial displacement of the sample crystal inside the furnace and the external mirror caused by the changes in thermal environment. When the crystal is partially transmitted at wavelength of the light source, Fabry–Perot interference is formed between the lights partially reflected from the front surface and the light reflected at the back surface of the crystal. The advantages of Fabry–Perot interferometry are (1) it does not need a calibration standard and (2) it only measures the relative spatial displacement of the front and back surfaces without any influence from the external environment. In this period, the thermal expansion of ZnSe was measured by Fabry–Perot interferometry from 17 to 1080 °C during the heating up of a ZnSe crystal, which was used as the seed for the following growth experiment by physical vapor transport.

Bibliography Type:	Description: (Last Updated: 02/18/2025)
Articles in Peer-reviewed Journals	Su C-H, Lehoczky SL, Li C, Knuteson D, Raghothamachar B, Dudley M, Szoke J, Barczy P. "Crystal growth of CdTe by gradient freeze in Universal Multizone Crystallizator (UMC)." Materials Science Forum. Solidification and Gravity IV 2006;508: 117–24. http://dx.doi.org/10.4028/www.scientific.net/MSF.508.117 , Mar-2006
Articles in Peer-reviewed Journals	Barczy P, Szoke J, Su C-H, Lehoczky SL, Schweizer M, Volz MP, Szofran FR. "Single crystal growth – new technologies on old base." Materials Science Forum, Solidification and Gravity IV, 2006;508:479-84. http://dx.doi.org/10.4028/www.scientific.net/MSF.508.479 , Mar-2006
Articles in Peer-reviewed Journals	Su C-H, Lehoczky SL, Raghothamachar B, Dudley M. "Crystal growth and characterization of CdTe grown by vertical gradient freeze." Materials Science and Engineering: B, 2008 Jan 25;147(1):35-42. http://dx.doi.org/10.1016/j.mseb.2007.11.005 , Jan-2008
Articles in Peer-reviewed Journals	Su C-H. "Energy band gap, intrinsic carrier concentration, and Fermi level of CdTe bulk crystal between 304 and 1067 K." Journal of Applied Physics. 2008 Apr 15;103(8):084903. http://dx.doi.org/10.1063/1.2899087 , Apr-2008
Articles in Peer-reviewed Journals	Su C-H, Feth S, Lehoczky SL. "Thermal expansion coefficient of ZnSe crystal between 17 and 1080 °C by interferometry." Materials Letters, 2009 Jul;63(17):1475-7. http://dx.doi.org/10.1016/j.matlet.2009.03.050 , Jul-2009
Articles in Peer-reviewed Journals	Su C-H, Lehoczky SL. "Melt growth of high-resistivity CdZnTe crystals by controlling Cd over-pressures." Journal of Crystal Growth, 2011 Mar 15;319(1):4-7. http://dx.doi.org/10.1016/j.jcrysgro.2010.11.177 , Mar-2011